Aurora Public Schools — Grade 6 Reading Achievement & Equity

Academic years 2022–23 and 2023–24

Cohort
Grade 6 assessed students

Observed windows
Fall 2022 to Winter 2023–24

Method
Transition & regression models

Independent analytical work sample; not an official APS publication.

Reading improvement is projected, but prior Spring benchmark may not be met

Winter 2023–24 observed

14.9%

Mid or Above · 328 / 2,207 students

Spring 2023–24 projected

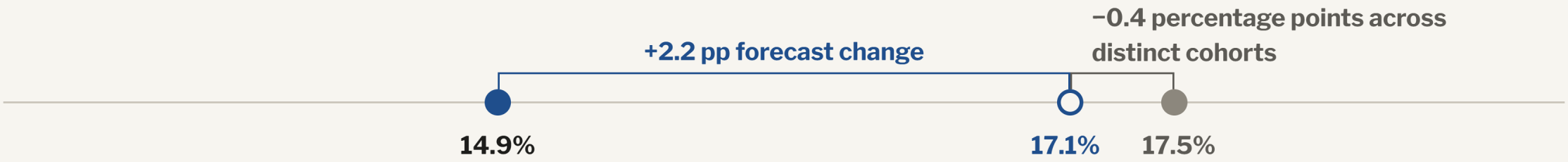
17.1%

Mid or Above · +2.2 pp forecast change

Spring 2022–23 prior benchmark

17.5%

Mid or Above · 361 / 2,059 students

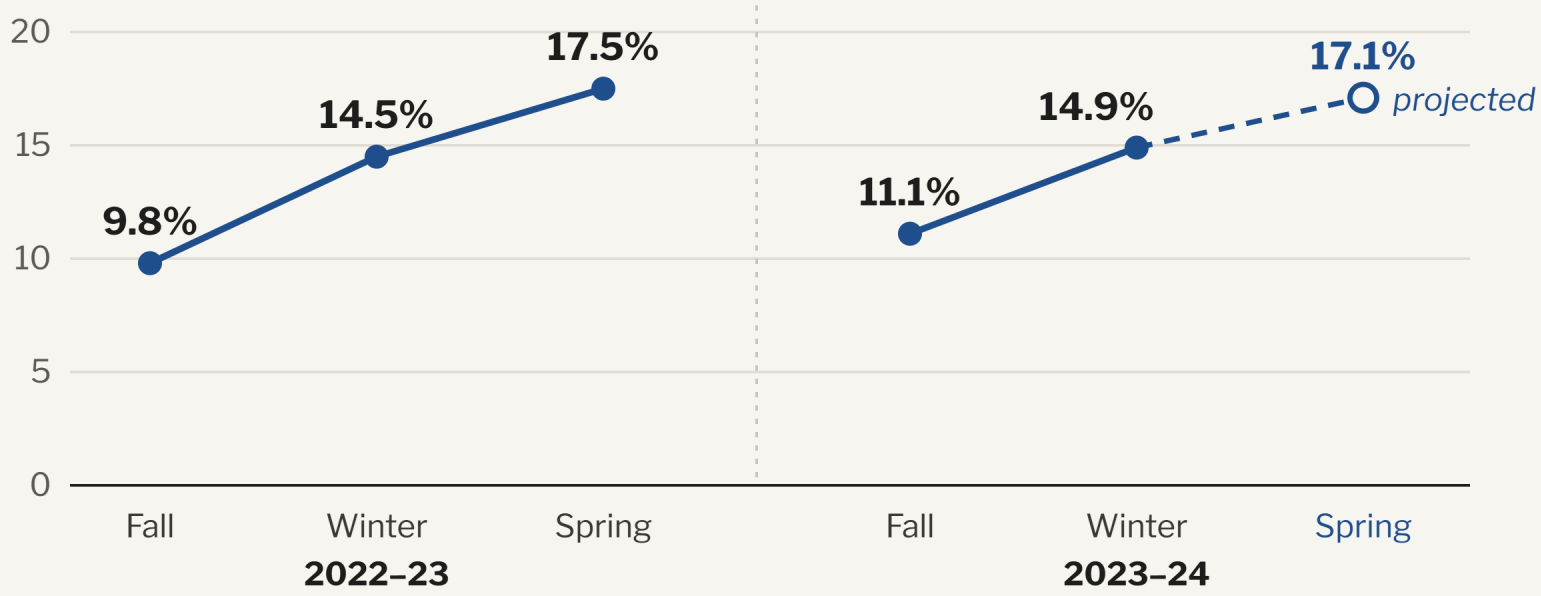


Note: Spring 2022–23 and projected Spring 2023–24 refer to different sixth-grade cohorts. The projection is a planning estimate, not an observed outcome or validated causal prediction.

Achievement rises across windows, with five observed periods

District achievement trajectory (2022–23 to 2023–24)

% Mid or Above



Winter-to-winter progress

+0.4 pp

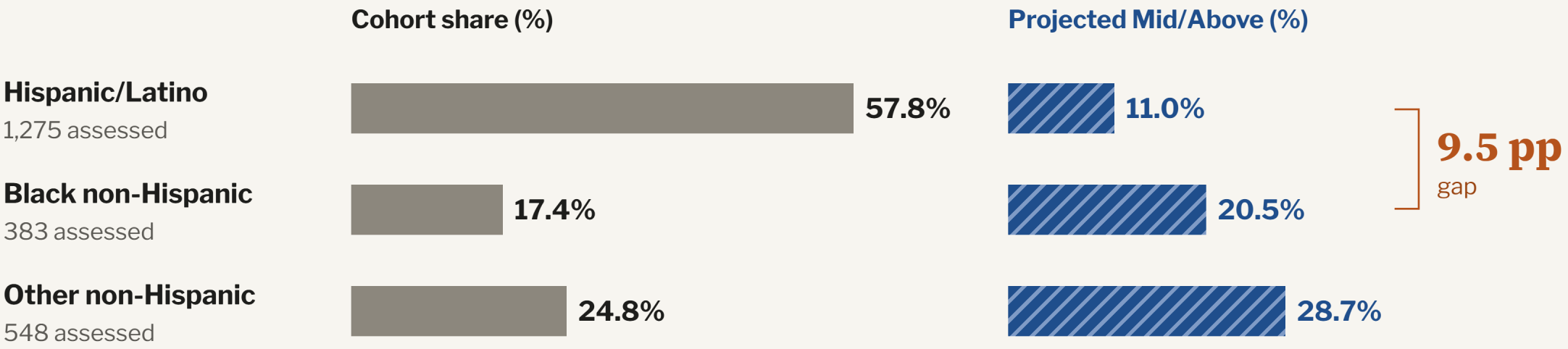
Growth from 14.5% (Winter 22–23) to 14.9% (Winter 23–24)

Observed

Projected

Note: Spring 2023–24 is a planning forecast; prior five windows are observed. The Winter-to-Winter change is a descriptive change, not a statistically established improvement.

Hispanic/Latino students form the largest assessed cohort



Projected difference between Black (20.5%) and Hispanic (11.0%) student groups: **9.5 pp**.

Note: All groups are assigned probabilities using the same district-pooled historical transition model. This is not evidence of an independent group-specific model or differences in school effectiveness.

Prioritize resources with assessed population counts

Top assessed Grade 6 middle schools (Winter 2023–24)

School	Assessed Grade 6 students	Hispanic/ Latino	Black non-Hispanic
Aurora Hills Middle School	**299**	182	59
Mrachek MS	**288**	156	49
East Middle School	**208**	159	26
Columbia Middle School	**201**	100	35
North MS	**184**	130	24

The summary contains 22 school/location categories, including highly specialized/small locations. Aurora Quest K–8 has 90 assessed students and 70.0% observed Winter Mid/Above; APS Avenues has 12 assessed students and 41.7% (5 of 12). A one-student change at APS Avenues changes its rate by approximately 8.3 pp.

Note: Do not treat counts as official enrollment.

Projected school trajectories vary across the district

Projected Spring 2023–24 change vs previous Spring (17 comparable schools)

School	Previous Spring 2022–23	Projected Spring 2023–24	Projected change vs previous Spring (percentage points)
Harmony Ridge PK–8	12.1%	**27.4%**	**+15.3 pp**
Aurora Quest K–8	71.1%	**61.7%**	**−9.4 pp**
Aurora Frontier K–8	40.0%	**35.4%**	**−4.6 pp**
Columbia Middle School	22.3%	**18.0%**	**−4.3 pp**

School projections differ; changes are not causal school-effect estimates.

Note: Only 17 of 22 school/location categories have a comparable prior-Spring result. Missing comparisons are not zero change. These comparisons involve different sixth-grade cohorts and pooled forecast probabilities.

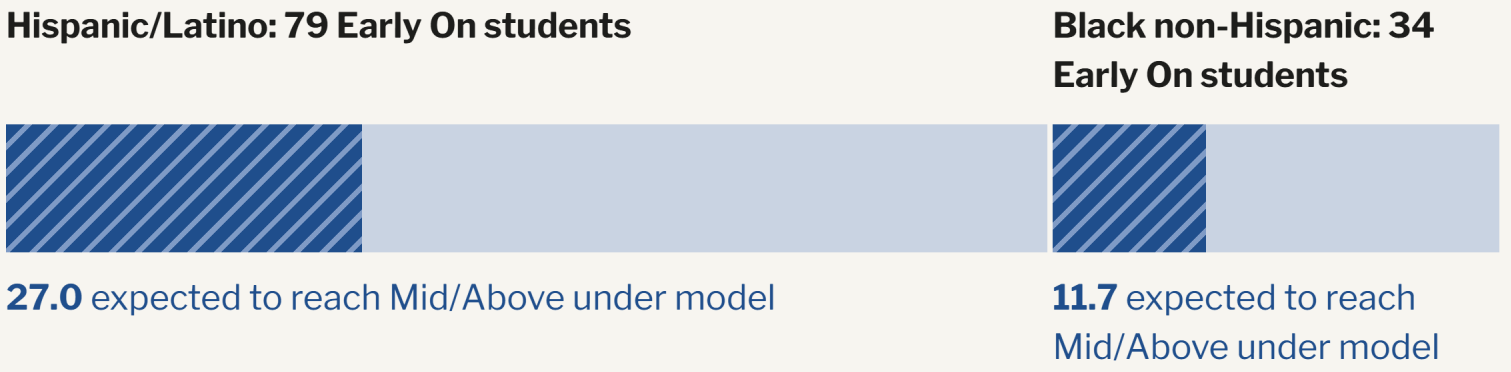
113 focus-group students are classified Early On Grade Level

Combined focus group, Winter 2023–24

113

Early On students

38.7 expected to reach Mid/Above under model



Where to investigate first: Columbia Middle School has 13 Hispanic/Latino Early On students; Aurora Hills has 10; Aurora Frontier and Charles Burrell each have 7.

"Early On Grade Level" is a placement category below Mid or Above Grade Level. It does not mean each student is a specified number of scale-score points from the cutoff. Expected outcomes are not estimated additional students produced by an intervention.

Note: Hatched area = expected to reach Mid/Above under model.

Sensitivity comparison between two forecast approaches

Model A 2,207 records scored

17.1%

Historical transitions approach

Match Winter and Spring 2022–23 Grade 6 assessments by student; estimate the observed Spring Mid/Above proportion within each Winter placement category; apply those proportions to Winter 2023–24 students.

Model B +0.5 pp over Model A

17.6%

Logistic regression approach

Predict Spring Mid/Above from the prior year's Winter scale score. Aggregating rounded school-level forecasts yields approximately 17.6% district-wide, about 0.5 pp above Model A.

Validation status: Independent held-out accuracy was not evaluated in console log.

Guidance: Use estimates for sensitivity comparison, not validated causal prediction.

Note: The available console log indicates the designated statistical-validation script was skipped, while the prediction-evaluation script's Brier calculation used training-set predictions. Do not label that score "validated performance."

Quality assurance confirms internal data consistency

Pipeline audit results across 46,135 matched assessment records

Assessment source records	**49,132**
Selected distinct assessment sets	**46,135**
Assessment records matched in final merge	**46,135 / 46,135**
Unmatched merged assessments	**0**

0

Quality failures

Across all reconciliation tests

2

Disclosure flags

Triggered for small cells

Audit conclusion: Internal consistency established; predictive validity remains unvalidated.

Note: The pipeline emitted CSV parsing warnings that should be investigated at their originating read_csv() calls. The two review flags concern small assessed school cohorts and small school-by-equity cells.

Five strategic decisions to direct support

Targeted action plan for district leadership

01	**Target need**	Allocate attention by cohort size (Aurora Hills, Mrachek, East).
02	**Monitor equity**	Review outcomes using both student counts and percentages.
03	**Screen near-benchmark**	Assess 113 Early On students for targeted skills.
04	**Investigate variance**	Review participation before assuming school-level declines.
05	**Validate models**	Test predictions against actual outcomes before operational use.

Essential disclosures and reporting safeguards

Guidelines for ethical interpretation and communication

Assessed cohort

Percentages reflect assessed Grade 6 records, not enrollment. Non-tested students are excluded from proficiency ratios.

Distinct cohorts

Prior Spring and projected Spring reflect different student groups across successive academic years.

Single forecast window

Only Spring 2023–24 is projected; 5 prior windows are observed longitudinal data.

Data governance

Suppress small subgroup cells; do not publish raw CSVs to protect student privacy and FERPA compliance.